



A comparison of iterative Fourier transform algorithms for image quality estimation

Dina Yaqoob Alsaka^{1,2} · Çağlar Arpali³ · Serap Altay Arpali¹

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Abstract

A comparison was established between two iterative Fourier transform algorithms (IFTAs), such as the original Gerchberg–Saxton (GS) and the mixed-region amplitude freedom (MRAF) algorithms, for the hologram reconstruction of different target images through the full reference image quality estimation (IQE) and pixel homogeneity in the Fourier plane presented theoretically and experimentally. The comparison was applied depending upon both algorithms based on a computer-generated hologram (CGH) implemented utilizing a reflective phase-modulated liquid-crystal spatial light modulator (LC-SLM) to obtain the digital kinoform holograms of the desired intensity distributions. These digital holograms were applied to reconstruct the intensity patterns for 852 nm, which represents a laser beam source. The theoretical and experimental results of the reconstructed patterns obtained using the MRAF algorithm were found to be smoother and better than the patterns obtained using the GS algorithm. Unmodulated light beam (dc term) is removed from the reconstructed patterns attributed to digital kinoform holograms of MRAF algorithm as an alternative to the theoretical and experimental results without using any additional optic equipment at the light path. Moreover, this paper discussed the full reference objective quality estimations, such as mean square error (MSE), peak signal-to-noise ratio (PSNR), structural content (SC), normalized absolute error (NAE), normalized cross correlation (NK), and homogeneity of pixels, through the contrast (Cont) and inverse difference moment (IDM) for numerical and experimental results. According to the two desired intensity distributions processed theoretically and experimentally, the results of MRAF algorithm were found to be in the highly accurate recovered phase, the quality of image was enhanced, and the dc term was decreased. Image quality estimation of full reference objective relay on the feedback algorithms experimental attestation has not been implemented yet.

Keywords Computer-generated hologram (CGH) · Kinoform · Iterative Fourier transform algorithms (IFTAs) · Optical hologram reconstruction · Diffractive optical element (DOE) · Full reference objective quality

1 Introduction

The amplitude and the phase information together might be intuitively needed to reconstruct an image. However, it becomes important to determine how to reconstruct the image depending upon the phase distribution only. The phase-recapture problem in a Fourier transform imaging system is generally addressed for finding a technique to reconstruct an image from partial knowledge in the input plane and the transform domains [1]. An ideal phase-recapture is a technique to obtain the retrieval of the phase from two intensity measurements on the input and the output planes in an imaging system [2–12]. The intensity distribution of optical fields is controlled using a computer-generated Fourier transform phase holograms known as kinoform. This would change the phase of the optical field at each pixel without

✉ Çağlar Arpali
c.arpali@cankaya.edu.tr

Dina Yaqoob Alsaka
201382600@student.cankaya.edu.tr; engdina28@yahoo.com

Serap Altay Arpali
saltay@cankaya.edu.tr

¹ Department of Electronic and Communication Engineering, Çankaya University, Yukarıyurtçu mah. Mimar Sinan cad. No: 4, 06790 Etimesgut, Ankara, Turkey

² Department of Laser and Optoelectronics Engineering, Nahrain University, jadriyah, Baghdad, Iraq

³ Department of Mechatronics Engineering, Çankaya University, Yukarıyurtçu mah. Mimar Sinan cad. No: 4, 06790 Etimesgut, Ankara, Turkey

attenuating the amplitude [13]. High-diffraction efficiency is achieved with the kinoform hologram-encoding method [14, 15]. The computer-generated hologram (CGH) physically is implemented utilizing a diffractive optical element (DOE), such as a pixelated spatial light modulator (SLM) [16, 17]. A pixelated SLM is a device employed to modulate the monochromatic light by encoding the pattern information from a computer and converting it into the required output patterns. In general, SLM has non-required characteristics because of the pixelated structure of a phase-only SLM, which is a typical issue when using a commercial SLM for holographic projection of the desired light patterns. These characteristics would give rise to the degradation of the reconstructed patterns quality. One of these characteristics is the unmodulated light beam, which is sometimes called the dc term. As the SLM type is applied in this paper, the dc term originates from the reflected light from the dead space zone (inactive area of SLM) between pixels; this dc term results in a strong, bright light spot at the center of the reconstruction. To overcome the problem of dc term, many researchers successfully used several methods to block it or shift the patterns from the center of the reconstruction plane [18–21]. In our paper, a technique is considered to reduce the dc term from the patterns that rely on the partitioning of the reconstruction plane and the determination of the guess initial phase added to the incident beam. CGH has special features to reconstruct the patterns of the desired distribution, such as: accuracy in matching the desired intensity distribution, smoothness of the profile generated by the CGH, and efficiency in diffracting light into the target profile [17, 22].

In general, from a required arbitrary intensity distribution in the output plane, the CGH cannot be directly computed. Thus, an iterative Fourier transform algorithm (IFTA) could be considered as a better means to compute a CGH compared to other methods, such as a direct binary search [23, 24].

For the purpose of our work, a phase-only SLM consisting of 256×256 programmable pixels is used; each pixel is able to impose phase retardation between 0 and 2π . The resulting irradiance distribution is obtained in the far field, or equivalently in the back focal plane of a lens. To the best of our knowledge, in the literature, an extensive comparison of Gerchberg–Saxton (GS) and mixed-region amplitude freedom (MRAF) algorithms for full reference of objective image quality estimation (IQE) based on an experimental demonstration has not been performed yet. In our paper, we implemented and compared the original Gerchberg–Saxton (GS) algorithm [25–27], which is the first practical iterative algorithm in the Fourier-phase-recapture process, with a new IFTA, which is called “mixed-region amplitude freedom” (MRAF) algorithm [17, 28, 29]. These two algorithms rely on a phase-modulated SLM to generate the optical kinoform holograms of the required intensity patterns’ distribution

inserted in the Fourier plane. The MRAF algorithm is used to minimize failures of the GS algorithm and achieve high-quality optically reconstructed patterns of the desired intensity distribution. This comparison is based on the objective image quality estimation (IQE) by measuring the full references denomination, such as: mean square error (MSE), peak signal-to-noise ratio (PSNR), structural content (SC), normalized absolute error (NAE), and normalized cross correlation (NK) and homogeneity of pixels from the contrast (Cont) and inverse difference moment (IDM) for the reconstructed patterns. In addition, the dc term reduction is considered as an important issue for holographic implementation in the optical image transform, encoding and processing techniques. Thus, in this paper, from theoretical and experimental results, the MRAF algorithm is found to be an efficient IFTA for reduction the dc term from the reconstructed patterns without using any suppressed method; without adding any additional optic components in the optical axis and better enhancement of the image quality than GS algorithm and the reference [30]. Finally, this comparison between the GS and MRAF algorithms demonstrates that the MRAF algorithm can be used for fields that require a regular output laser beam in processing, such as optical communications, medical applications, industry applications, data storage, and optical metrology.

2 Algorithms to calculate CGHs

Computer-generated hologram (CGH) is considered a more flexible process that does not require the physical measurements of the object, i.e., only a mathematical depiction of the object is required. When physical recording of the complex wave fronts is used [31], in general, values of two arrays are used to describe the amplitude and phase of the diffracted wave from the imaginary object. The real definition of the object is given, and random phase is generated across the hologram plane to achieve the best quality [32]. To design and obtain the phase profile of any desired light distribution, the iterative Fourier transform algorithm (IFTA) must be utilized [33]. Figure 1 shows the basic concept of the computer-generated hologram (CGH) using the Fourier transform. In this figure, the SLM plane is assumed as the input plane, which contains the information of the diffracted patterns at the image plane. Two types of algorithms are intended to be discussed in this paper, the Gerchberg–Saxton (GS) and mixed-region amplitude freedom (MRAF) algorithms. MATLAB version R2013a was utilized to write these codes.

2.1 Gerchberg–Saxton (GS) algorithm

The GS iterative optimization algorithm is a phase-recapture algorithm that is classified in the IFTA group [27, 33, 34].

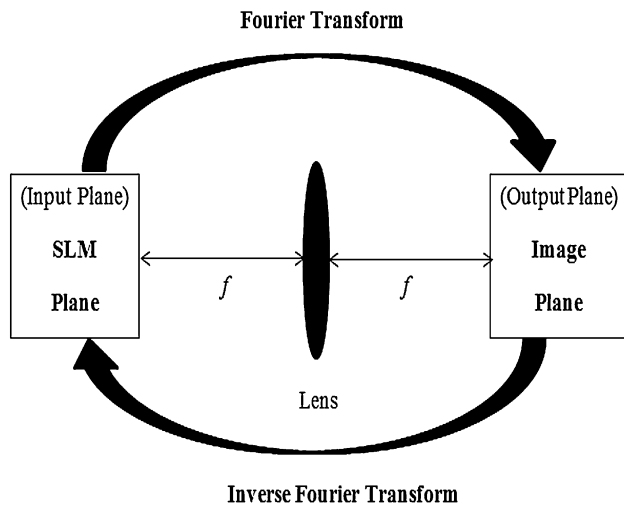


Fig. 1 CGH sequences

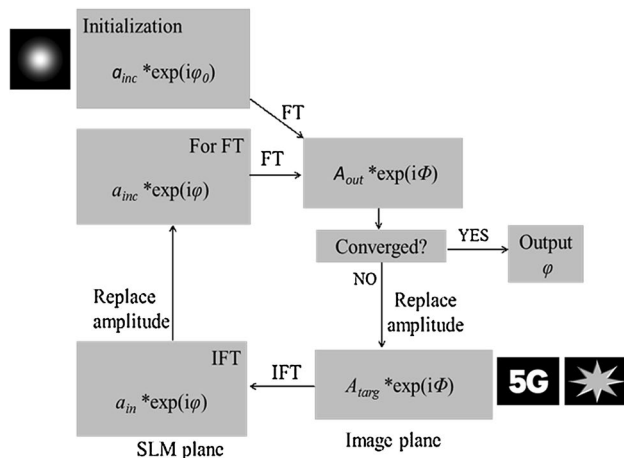


Fig. 2 Gerchberg–Saxton (GS) algorithm

Figure 2 sketches the principal steps of the GS algorithm. Before starting the algorithm, a random phase mask is added to the desired intensity distribution to accelerate the convergence of the algorithm [13, 35, 36]. This algorithm is initialized by creating a complex field from the amplitude of the incident light (a_{inc}) which is usually chosen to be uniform over the whole hologram and an initial guess for the phases (φ_0) as a random phase or uniform phase. Fourier transform (FT) is applied to this field $a_{inc} \exp(i\varphi_0)$ to yield $A_{out} \exp(i\Phi)$ [27]:

$$A_{out} \exp(i\Phi) = F(a_{inc} \exp(i\varphi)), \quad (1)$$

where A_{out} corresponds to the amplitude of the light field that would form in Fourier plane and is called the propagated field. Now, the algorithm process can be executed. For each point in A_{out} , the difference from the corresponding point of

the target, A_{targ} , was calculated, and the root-mean squared error was computed. If this value lies below a specific value, which must be defined by the user, then the algorithm is considered converged and φ is the final kinoform phase hologram of the desired intensity distribution [33, 37]. The root-mean squared error is given by

$$RMSE = \frac{1}{\sqrt{N}} \sqrt{\sum_{i,j} (A_{out}(i,j) - A_{targ}(i,j))^2}, \quad (2)$$

where i and j indices are the points in the sampled fields, and N is the number of pixels in the SLM plane. If the algorithm is not converged, then the next step is to replace the amplitude of the output A_{out} by the amplitude corresponding to the target pattern A_{targ} , while the calculated phases are kept [27, 33]:

$$A_{out} \exp(i\Phi) \rightarrow A_{targ} \exp(i\Phi). \quad (3)$$

In the next step, the inverse Fourier transform (IFT) of the above expression is taken

$$a_{in} \exp(i\varphi) = F^{-1}(A_{targ} \exp(i\Phi)), \quad (4)$$

where F^{-1} is the operator of the inverse Fourier transform. Finally, a_{in} is replaced by the incident amplitude a_{inc} and the next iteration starts as

$$a_{in} \exp(i\varphi) \rightarrow a_{inc} \exp(i\varphi). \quad (5)$$

In reality, the algorithm propagates the field back and forth between the hologram (input) and the image (Fourier or output) planes, as indicated in Fig. 1, using the amplitudes of the incident light and of the desired distribution as restrictions on the development of the phases [33, 37]. From Eq. (4), its argument is used to obtain the final kinoform phase that is loaded to the SLM, which is considered to be irradiated by the plane wave distribution [35].

Note that convergence of the GS algorithm is extremely erratic because of constraints on the amplitudes in process planes. Thus, the GS algorithm is improved to the MRAF algorithm to overcome these constraints.

2.2 Mixed-region amplitude freedom (MRAF) algorithm

The MRAF algorithm is an improved version of the Gerchberg–Saxton (GS) algorithm and it is effective in solving the general amplitude-phase-retrieval problem by relaxing the restriction on the amplitudes in the Fourier plane [29]. This is achieved by partitioning the Fourier plane into two parts: a signal region (SR) and a noise region (NR), where (SR) can be defined as all points of the target intensity distribution with non-zero values, and (NR) is referred to the zero

potential in target distribution [37]. The effect of dividing the Fourier plane into subsets results in a convergence of the algorithm very close to the target profile within (SR) and it reduces the dead space zones, which are caused the appearance of dc term in the reconstructed patterns, while the behavior in (NR) falls under less control. The relative distribution of the optical power in the two regions is quantified by the mixing parameter m ; $0 \leq m \leq 1$ [17]. During the iteration, the amplitude A_{out} is replaced by A_{targ} only for the pixels in (SR). In other words, the usable area of the Fourier plane is reduced compared to the GS algorithm [33, 37]. Accordingly, Eq. (3) becomes

$$A_{\text{out}} \exp(i\Phi) \rightarrow (|mA_{\text{out}}|_{\text{SR}} + |(1-m)A_{\text{out}}|_{\text{NR}}) \exp[i\Phi], \quad (6)$$

where $|\cdot|$ is the absolute value. Equation (6) is called the mixed field (E_{mix}), which is generated from the combination of the target and the propagated fields [29]. Therefore, the insertion of the target amplitude pattern in the mixed field is a major step to obtain the diffracted field intensity that approaches the desired pattern for each iteration [28]. In the input plane, a creation of a new field can be established by taking the inverse Fourier transform (IFT) of Eq. (6) as

$$A_i^{(n+1)} = |a_{\text{inc}}| \exp \left[i \arg \left[F^{-1} \left[E_{\text{mix}}^{(n)} \right] \right] \right], \quad (7)$$

where F^{-1} is the operator of the inverse Fourier transform. The required kinoform hologram sent to the SLM is equal to the argument of Eq. (7).

Comparing the GS and MRAF algorithms, the initial phase of MRAF is chosen to be close to the desired intensity distribution, which is considered as the method to generate a new light field at the output plane, unlike the GS algorithm, which was chosen in a random phase. The initial phase will be explained in the next section. The MRAF algorithm has an improved field distribution to ensure that the output plane (Fourier plane) does not include optical vortices, points characterized by a phase singularity and zero intensity, whereas the GS algorithm is not able to eliminate those issues [38, 39]. A block diagram of the MRAF algorithm is shown in Fig. 3.

2.3 Initial phase of algorithms

An initial guess phase produces a result that is close to the target distribution to improve the convergence rate, thereby reducing the number of iterations. Because the IFTAs being used when the solution of the CGH problem are unknown, choosing an initial phase profile φ to reproduce complex features in A_{out} is not possible. Thus, we must find an appropriate φ_0 as the starting point that contains most of the power in the output plane that roughly

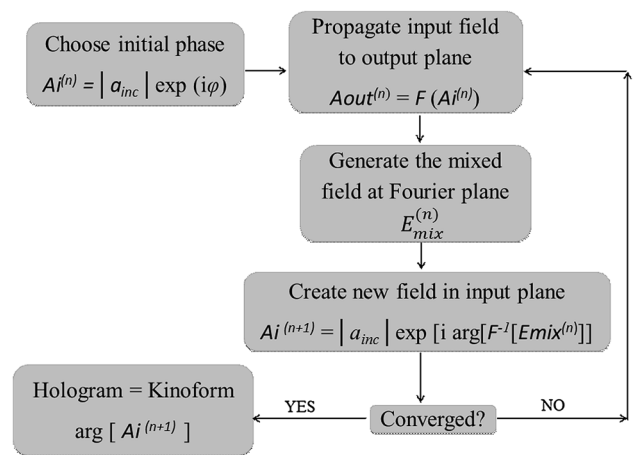


Fig. 3 Block diagram of MRAF algorithm

overlaps with the envelope of target intensity. The distribution of the initial phase must be chosen to prevent the appearance of any undesired vortices in the first propagated field. This task is achieved by adding zero intensity regions to the optical field to become larger than the size of CGH [40]. A quadratic phase profile combined with linear gradients distributions is suitable to ensure that the hologram phase is close to the desired intensity patterns at the Fourier plane [41]. A quadratic phase profile can be expressed as [42]

$$\varphi(x, y) = \frac{k}{2f}(x^2 + y^2), \quad (8)$$

where k is the wave number, and f is the focal length. Equation (8) is used to change the size of the envelope of the intensity profile in the Fourier plane. A linear gradient phase profile is [42]

$$\varphi(x, y) = 2\pi(G_x x + G_y y), \quad (9)$$

where G_x and G_y are the frequencies in x and y directions, respectively [42]. Equation (9) is used to shift the centroid of the intensity profile in the output plane to avoid complications caused by unmodulated light beam (dc term) resulting from the finite efficiency of a physically implemented CGH. Quadratic and linear phase distributions are added together modulo 2π to combine the effects of each initial phase distributions. This combination of the initial phase distributions has enough flexibility to obtain sufficient overlap between the first propagated and the reconstructed patterns. In this paper, the initial phase is assumed with the MRAF algorithm only, and the GS algorithm is used a random phase mask. In the next section, we will discuss the arbitrary intensity pattern shaping using a phase-only spatial light modulator (SLM).

3 Arbitrary intensity pattern shaping with phase-only SLM

The spreading of the reflected intensity pattern by the SLM is affected by the modulation over the phase freedom. Based on the interference between different parts of the beam pattern, the intensity profiles after various propagation distances will be changed. At a certain distance after the SLM, a specific target intensity pattern can be attained [29]. The SLM is fixed in front of the lens at its focal length f to define the input plane. The reconstructed pattern of intensity distribution is located at the back focal plane of the lens (image plane), as shown in Fig. 1. The electric field at image plane is given by the exact Fourier transform (FT) of the electric field at the input plane [43, 44]:

$$E_{\text{FP}}(x, y) = \frac{1}{i\lambda f} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E_{\text{IP}}(X, Y) e^{-2\pi i \frac{xX}{\lambda f}} e^{-2\pi i \frac{yY}{\lambda f}} dXdY, \quad (10)$$

where X, Y and x, y are the spatial and frequency coordinates of the input and output planes, respectively. The intensity pattern at the input plane is modeled as Gaussian beam intensity before it is modulated by the SLM. Thus, a Gaussian electric field distribution can be presented as [44]

$$E_{\text{G}}(x, y) = A_0 \exp\left(\frac{-(x^2 + y^2)}{w^2}\right), \quad (11)$$

where A_0 is the magnitude constant, and w is the beam waist, which is assumed to be symmetric in horizontal and vertical directions. For a beam modulated by phase-only SLM, the modulated field at the input plane E_{IP} has a restricted shape, where the amplitude is given by a broad Gaussian distribution of the laser with a nearly plane wave distribution that is achieved using a collimation unit, which will be described in optical setup and “Results” section; the phase modulation of the SLM is the initial step of IFTAs:

$$E_{\text{IP}}(x, y) = E_{\text{G}}(x, y) \exp(i\varphi_{\text{SLM}}). \quad (12)$$

The first propagated field to the output plane is given by

$$E_{\text{FP}}(x, y) = F[E_{\text{IP}}(x, y)], \quad (13)$$

where F as the operator of Fourier transform. The phase guess in the next iteration step is defined as the phase component of the inverse Fourier transform (IFT) of Eqs. (3 and 6) for GS and MRAF algorithms [29]:

$$\varphi_{\text{SLM}}^{(n+1)} = \arg(F^{-1}(A_{\text{out}})). \quad (14)$$

Because the algorithms will be recursive for n th iterations to match with the target distribution at the output plane as E_{FP}^n . Equations (3 and 6) are used to obtain the reconstructed patterns of the required intensity distribution and the final kinoform hologram of E_{FP}^n that was calculated from Eq. (14)

and loaded to the SLM as the CGH of the target patterns. In the next sections, we will discuss the image quality estimation (IQE) and the homogeneity of pixels.

4 Image quality estimation (IQE)

In general, there are many researchers dealing with image quality using different techniques which are based on creating a deterministic or random phase mask coding [30, 45, 46]. In our work, we are focusing on the two IFTAs such as GS and MRAF algorithms based on the SLM device to determine the kinoform hologram by dividing the Fourier plane into two different regions with MRAF algorithm according to the non-zero intensity value (SR) and the zero intensity value (NR) to describe the desired intensity distribution which inserted in the Fourier plane. When comparing our work with reference [30], the image quality estimation is based on partitioning of Fourier plane into two regions (SR and NR) which are variable regions and depends on the intensity distribution of the desired shape inserted in Fourier plane meaning that the shape and size of these regions are restricted by the bright and dark regions of the desired shape; thus, at the input plane, there is no partitioning of the intensity distribution only a plane wavefront distribution is assumed, while in reference [30], these regions are called signal and dummy regions which are created by random data phase mask made by the user to describe the incident light intensity which means that these regions are not related to the profile inserted in Fourier plane. In addition, in reference [30], there is no change between the modulus of the propagated (Fourier transform of the input field) and desired fields in these two regions at the Fourier plane to obtain the hologram with high quality, as well as the convergence condition is satisfied by comparing between the field of the inverse Fourier transform and the input field, whereas in our work, this condition is based on comparing between the fields of desired distribution and the formed distribution at the Fourier plane. In this study, the kinoform hologram of the desired intensity distribution is determined utilizing the MRAF algorithm as a new version of iterative Fourier transform algorithms which is based on the replacement of the modulus of these fields from (SR) only at the Fourier plane. Depending on the partitioning of MRAF algorithm and the determination of the initial phase from the desired intensity distribution the dc term in the reconstructed image is successfully eliminated without using any additional filter or inserted optical component in the experimental setup or moving the detector plane, while this dc term still appeared in reference [30, 45, 46] and the image quality estimation presented in this study is related to the full reference objective quality measurements only. In addition, the number of iterations required to obtain the kinoform hologram which

is loaded into the SLM device is reduced when compared with the reference [30]. Thus, this work has not yet been submitted in the field of the full reference objective quality measurements using SLM device to reconstruct patterns.

The estimation of image quality can be classified usually into two categories: subjective and objective quality measurements. Subjective quality measurements are the result of human experts providing their opinion of the image quality and it is usually too inconvenient, time-consuming, and expensive. In contrast, the objective quality measurements are implemented with mathematical algorithms without human involvement. Such methods could eliminate the need for expensive subjective studies. In this study, the focus is on the full reference objective quality measurements using SLM device to reconstruct patterns based on the digital hologram kinoform. The desired patterns as the reference images have been compared with decoded images (reconstructed patterns) obtained using the GS and MRAF algorithms. In addition, the homogeneity of pixels using a gray-level co-occurrence matrix (GLCM) is discussed, because the reconstructed patterns of the desired distribution by SLM are considered as gray-scale images. In the next sections, the full reference objective quality measurements and pixel homogeneity will be explained in detail.

4.1 Full reference objective quality estimation

First, the reference image (desired patterns) and the decoded image (reconstructed patterns) are taken as $r(m, n)$ and $R(m, n)$, respectively. M and N are the numbers of pixels in x and y directions [47–50]. Images are taken as the gray-level scale in all analyses.

4.1.1 Mean square error

The mean squared error (MSE) is the simplest criterion of image accuracy measure. The purpose of image accuracy measure is to provide a quantitative result that described the degree of conformity, i.e., the level of deformation between the reference and the decoded images. MSE is given by

$$\text{MSE} = \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N (r(m, n) - R(m, n))^2. \quad (15)$$

The large value of MSE, according to Eq. (15), indicates that the image has a poor quality and the range of values of MSE is $[0, \infty]$.

4.1.2 Peak signal-to-noise ratio

The peak signal-to-noise ratio is the ratio between the extreme value (power) of a signal and the power of deforming noise that affects the quality of its representation; it is measured in dB. The peak signal-to-noise ratio is denoted

by PSNR; a small value indicates that the image has a bad quality. PSNR is given by:

$$\text{PSNR} = 10 \log_{10} \frac{(2^n - 1)^2}{\sqrt{\text{MSE}}}, \quad (16)$$

where n is the bit of information in image; the maximum value is 255 and the minimum value is 0 for gray-level images.

4.1.3 Structural content

Structural content (SC) is a global measure that compares the total weight of the reference and the decoded images. The expected value of SC is 1. SC is defined as

$$\text{SC} = \frac{\sum_{m=1}^M \sum_{n=1}^N r(m, n)^2}{\sum_{m=1}^M \sum_{n=1}^N R(m, n)^2}. \quad (17)$$

4.1.4 Normalized absolute error

Normalized absolute error (NAE) is defined as the ratio of the difference between the reference and decoded images to the reference image. A large value of NAE indicates that image is of poor quality, whereas a value approaching zero indicates that the image has a better quality. NAE is defined as follows:

$$\text{NAE} = \frac{\sum_{m=1}^M \sum_{n=1}^N |r(m, n) - R(m, n)|}{\sum_{m=1}^M \sum_{n=1}^N |r(m, n)|}. \quad (18)$$

4.1.5 Normalized cross correlation

The normalized cross correlation is defined as the ratio of sum of the multiplication of reference and the decoded images to the square sum of reference image; it is denoted by NK and is defined as

$$\text{NK} = \frac{\sum_{m=1}^M \sum_{n=1}^N r(m, n) \times R(m, n)}{\sum_{m=1}^M \sum_{n=1}^N r(m, n)^2}. \quad (19)$$

If the difference between the reference and decoded images is zero, then the NK value tends to 1; i.e., NK gives the convergence of images.

4.1.6 Pixels homogeneity

In image-processing analyses, the homogeneity of an image normally refers to change in pixel intensity values. In this paper, the gray-level co-occurrence matrix (GLCM) technique is used to measure the contrast and inverse difference moment (Homogeneity) for the numerical and experimental

results of the reconstructed patterns of the desired intensity distribution. GLCM is based on the extraction of a gray-scale image, which depends on the relationship between two neighboring pixels [51]. GLCM is a square matrix with N dimensions.

4.1.7 Contrast

Contrast is the change in gray-level or a measure of intensity between the pixels over the whole image. The contrast is 0 for a constant gray-level image. The contrast is denoted by $Cont$ and is given by [51]

$$cont = \sum_{k=0}^{N-1} k^2 \left\{ \sum_{i=1}^N \sum_{j=1}^N P(i,j) \right\}, |i,j| = k, \quad (20)$$

where $P(i,j)$ is $N \times N$ GLCM, and i,j are the numbers of occurrence of pixels.

4.1.8 Inverse difference moment

Inverse difference moment (IDM), sometimes call homogeneity, measures the homogeneity of a digital image. IDM involves the measuring of the proximity of the distribution of the GLCM elements to the GLCM diagonal. In general, homogeneity value ranges from a low value of 0 to high value as 1; thus, the quality of image is very close to the regular structure and it is defined as [52]

$$IDM = \sum_{i=1}^N \sum_{j=1}^N \frac{P(i,j)}{1 + (i-j)^2}. \quad (21)$$

In this paper, MATLAB codes for measuring the image quality and pixels homogeneity were used as effective programs for image quality estimation.

5 Simulation test of the IFTA algorithms

A numerical simulation to test the effectiveness of IFTAs was conducted in this paper. This simulation was performed based on Eq. (11). As an input to Eq. (11), a laser beam with a large diameter of 20 mm with an assumed uniform plane wave distribution was used to irradiate the physical surface of the SLM device. This uniform plane wave was implemented experimentally using a collimation unit, as explained in optical setup and results' section. The next step in this simulation is to define the desired pattern distribution at the Fourier plane; two digital images as the desired intensity distribution are chosen with different bit depths and size of pixels. These two desired distributions are 5G and star shapes of 8-bit color stored per channel. These images are considered as gray-scale images of bitmap file type used in this process with respect to signal and noise regions and are shown in Fig. 4. These regions are variable with respect to the desired intensity distributions.

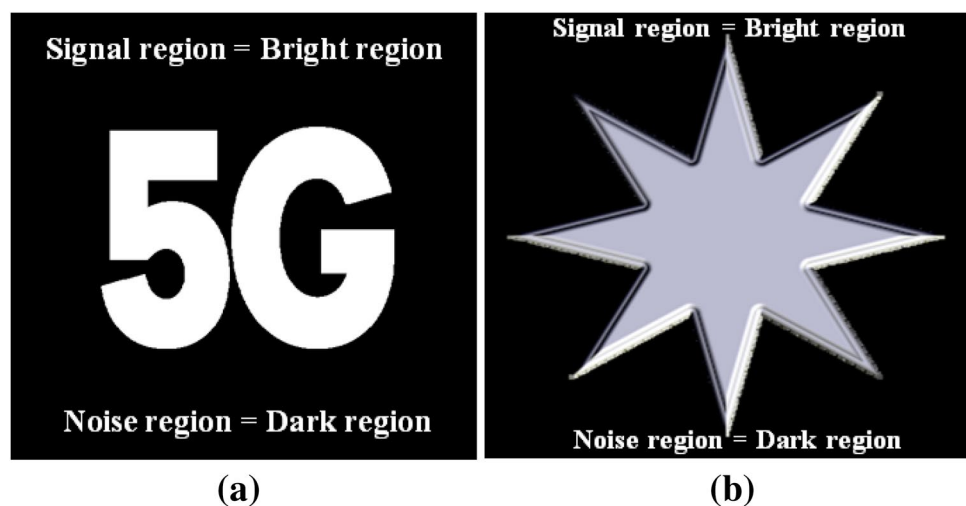
After 100 iterations of Eqs. (3, 6, and 14) for both algorithms and using a lens of focal length 250 mm, the desired patterns of intensity distribution are reconstructed in the far field. A CCD camera is placed at a distance equal to the focal length to record the optically reconstructed patterns.

6 Results and discussion

6.1 Numerical results and discussion of GS algorithm

In the GS algorithm, the initial phase of the target patterns was assumed as a random phase of 512×512 pixels that depends on the size of the target. Equations (3 and 14) are used to compute the phase kinoform loaded to the SLM to

Fig. 4 Target patterns as bitmap file with **a** 5G shape and **b** star shape



form the reconstructed patterns. The reconstructed patterns of the desired distribution are formed at Fourier plane when the plane wave distribution is incident on the kinoform hologram at the SLM plane. The results of phase kinoform hologram determined digitally and the reconstructed patterns for two desired distributions are shown in Figs. 5 and 6, respectively.

Root-mean square error (RMS %) value, which depends on Eq. (2), of the reconstructed patterns in Fig. 6 are 0.2185 and 0.1743 for 5G and star shapes, respectively. Figure 6 shows that the reconstructed profiles at the output plane are matched very well with the desired patterns, but it includes distortion (noise), i.e., these patterns have poor smoothness. A bright region in the middle of the reconstructed patterns is caused by the unmodulated light beam (dc term). Thus, to enhance the quality of the reconstructed patterns of the desired distribution, the modified GS algorithm (MRAF)

was used in this paper by reducing the distortion to its lowest levels and reducing the unmodulated light beam (dc term).

6.2 Numerical results and discussion of MRAF algorithm

The same considerations of the GS algorithm were utilized with another appropriate initial phase computed from the combination of Eqs. (8 and 9), and with an empirical value of the mixed parameter $m=0.4$ used in Eq. (6). Equations (6 and 14) with a plane wave distribution are utilized in the MRAF algorithm to calculate the kinoform hologram and the reconstructed patterns. The results of the final kinoform hologram and the reconstructed patterns of the desired distribution are shown in Figs. 7 and 8.

The root-mean square error (RMS%) value for the reconstructed patterns is 0.0447 and 0.0462 for 5G and

Fig. 5 Kinoform hologram of GS algorithm for 5G shape (a) and star shape (b)

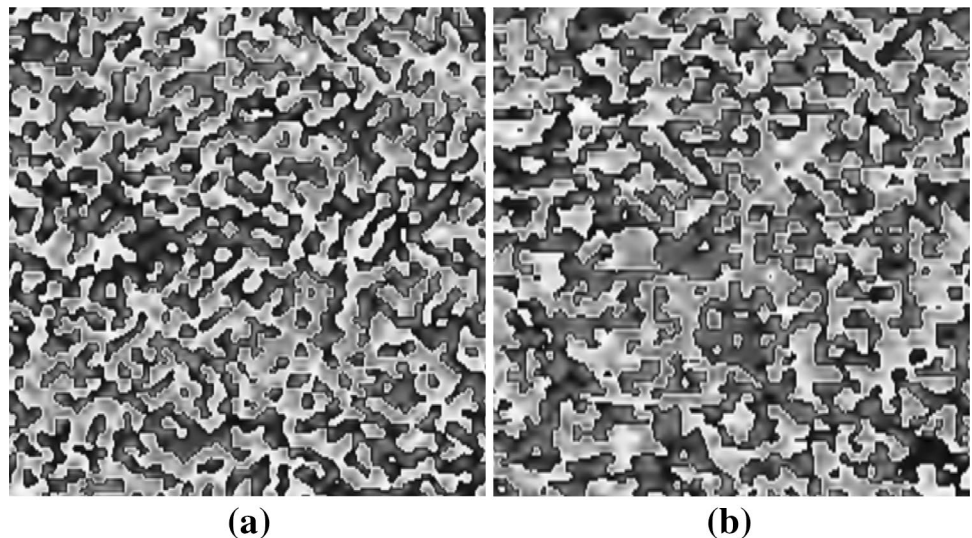


Fig. 6 Reconstructed patterns of GS algorithm for 5G shape (a) and star shape (b)

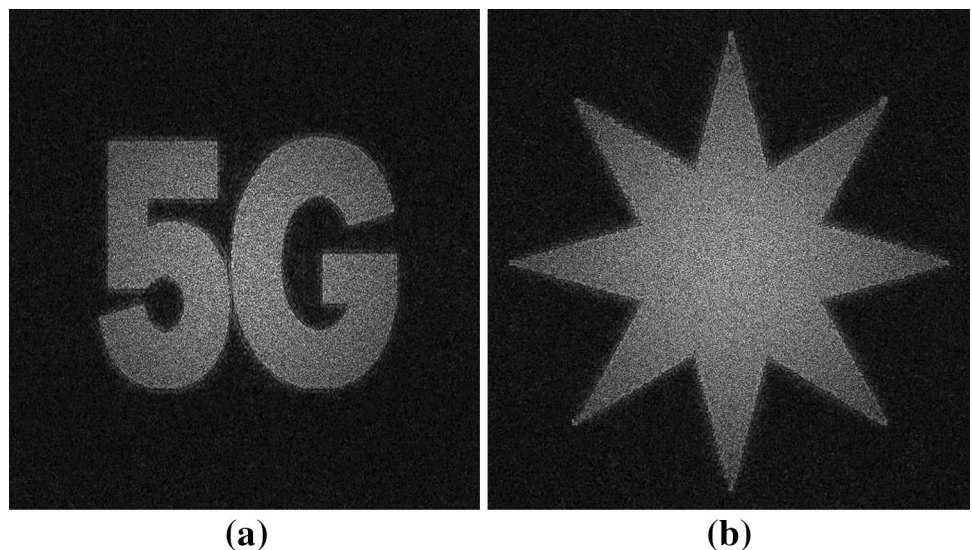


Fig. 7 Final kinoform holograms of MRAF algorithm for 5G shape (a) and star shape (b)

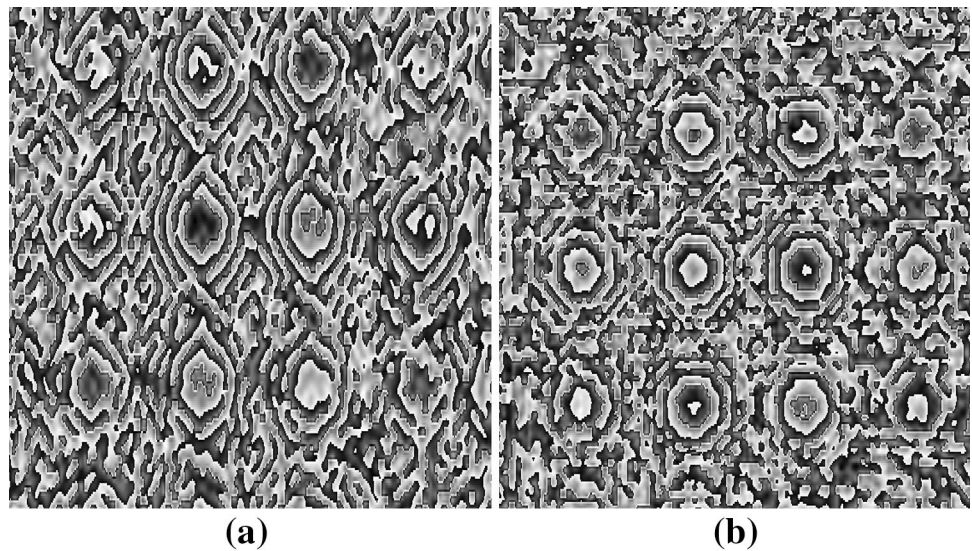
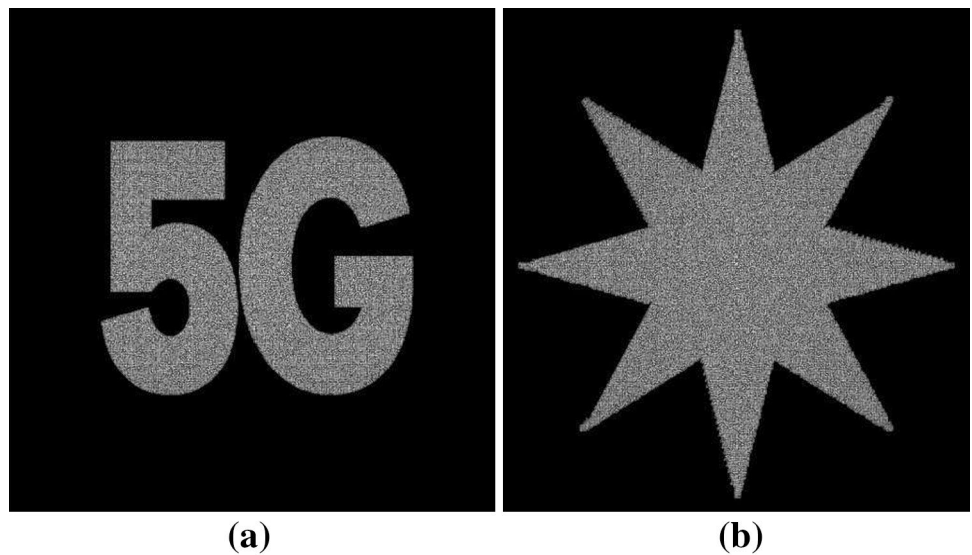


Fig. 8 Reconstructed patterns of MRAF algorithm for 5G shape (a) and star shape (b)



star shapes, respectively. The RMS values with MRAF algorithm are very small compared with those of the GS algorithm because of the choice of the initial guess phase and the partitions of Fourier plane. When comparing Figs. 6 and 8, the optically reconstructed patterns have an acceptable uniformity, better smoothness and the dc term is reduced or removed with MRAF algorithm without using any compressed method in algorithm. This algorithm splits the Fourier plane into two regions, thereby reducing the dead space zones between pixels in the SLM. These enhancements of the optically reconstructed patterns of the desired intensity distribution of the MRAF algorithm are clearly observed in the practical results.

6.3 Numerical results of IQE and discussion

Optically reconstructed patterns of the desired intensity distribution from GS and MRAF algorithms and Eqs. (15–21) are used to measure the full reference objective quality parameters and the homogeneity of pixels. Table 1 shows the measured values of IQE and pixels homogeneity of the numerical reconstructed patterns for GS and MRAF algorithms.

From Table 1, low values of contrast and high values of IDM between the desired and the reconstructed patterns of intensity distribution are obtained with the MRAF algorithm compared to those obtained using the GS algorithm. The SC values are very close to the expected

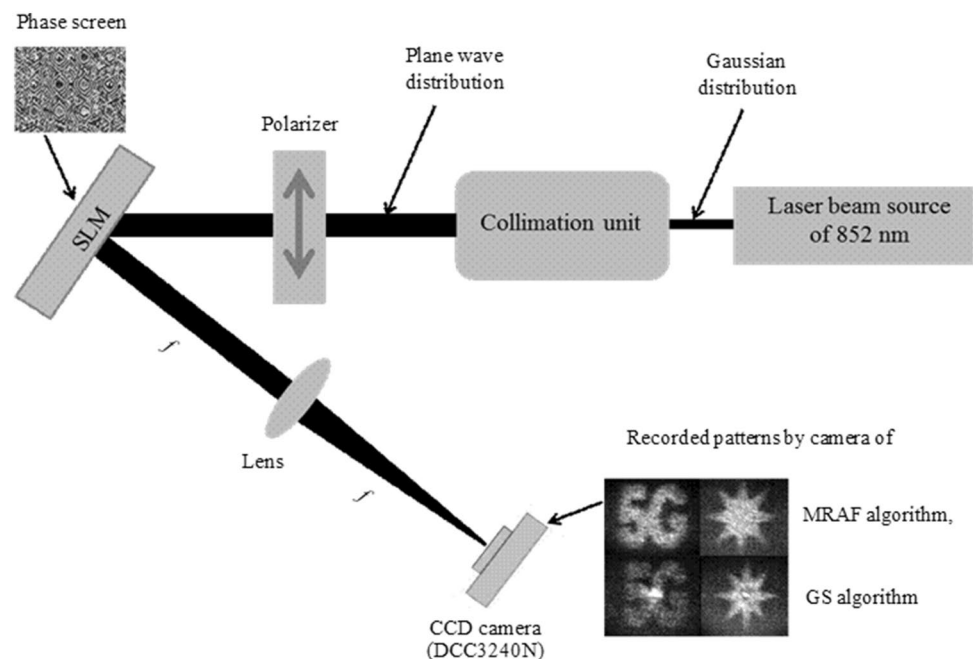
Table 1 Measured values of image quality for the numerical reconstructed patterns

Algorithm	Target patterns	MSE	PSNR	SC	NAE	NK	Cont	IDM
GS	5G shape	51.8531	30.9831	0.2889	0.5966	1.0024	0.5159	0.8870
	Star shape	61.3033	30.2560	0.3613	0.4101	1.0154	0.7159	0.8393
MRAF	5G shape	51.6178	31.0028	0.9350	0.4992	1.0011	0.3789	0.9151
	Star shape	57.9122	30.5031	0.9197	0.3333	1.0147	0.4741	0.8919

value as 1 for the reconstructed patterns of the MRAF algorithm, whereas the SC values are small and far away from the expected value for the reconstructed patterns of the GS algorithm. The PSNR values of the patterns of the MRAF algorithm are greater than the PSNR values of the GS algorithm by a small amount. In addition, the NAE values of the reconstructed patterns of the MRAF algorithm are small when compared with those values of the GS algorithm. The other parameters in Table 1 also indicate that the results of the reconstructed patterns of intensity distribution obtained by MRAF algorithm are better than the results obtained by GS algorithm. Thus, the reconstructed patterns of the desired distribution are in acceptable level of smoothness and homogeneity, and the image quality is high and the distortion of the reconstructed patterns is removed using MRAF algorithm to calculate the kinoform hologram in field of computer-generated hologram (CGH). The MRAF algorithm shows a high quality for the reconstructed patterns of the desired distribution in the Fourier plane.

6.4 Optic setup and results

The transformation of the desired shape is experimentally performed through the use of a phase-only SLM. A laser source emitting a Gaussian beam of a wavelength of 852 nm was considered. The Gaussian beam is transferred and expanded to beam of a flat wave front with diameter of 20 mm using a collimation unit. The collimation unit consists of a beam expander, a pinhole, and a biconvex lens of a focal length 175 mm that are used to obtain a flat distribution to fill the active area of SLM after passing through a polarizer. A polarizer is utilized to control the phase modulation of light field incident on the SLM. In this paper, the SLM device is a (Holoeye-Pluto based on reflective LCOS microdisplays of 1920×1080 -pixel resolution with a small pixel pitch of $8 \mu\text{m}$ and a reflectivity of 62% for used wavelength). The plane of the SLM is Fourier transformed with a biconvex lens of focal length $f = 250 \text{ mm}$. The reconstructed patterns of the desired distribution are recorded as gray-scale images using a digital CCD camera (DCC3240N) of 1280×1024 pixel resolutions placed at the Fourier plane, as shown in Fig. 9.

Fig. 9 Optic setup

To compare the optical performances of the reconstructed holographic patterns utilizing the GS and MRAF algorithms, experiments are performed. Based on the optical setup that provides the required flat wave front distribution, the final kinoform holograms are displayed in Figs. 5 and 7. The reconstructed patterns of the desired distribution for both algorithms are recorded at the output plane using a digital CCD camera. The recorded patterns of GS and MRAF algorithms for two desired distributions are shown in Figs. 10 and 11, respectively.

By comparing Figs. 10 and 11, the recorded patterns of the desired distribution obtained by the MRAF algorithm are found to be better than these patterns obtained by the GS algorithm. The reconstructed patterns obtained by the MRAF algorithm are more uniformly smooth and do not have a dc term in patterns, whereas the optically

reconstructed patterns obtained by the GS algorithm are somewhat deformed and have a dc term. The appearance of the dc term in the reconstructed patterns of the GS algorithm with that strong bright form is coming from both a large dead space area in the reconstruction plane and the low reflectivity of SLM for the applied wavelength (approximately 62%) as well. Dc term is reduced from the reconstructed patterns with holograms of MRAF algorithm without using any additional optic instruments (such as special filter placed in the light path that would decrease the efficiency of the entire system). Therefore, the MRAF algorithm is more suitable to generate the kinoform holograms of the desired distribution and to enhance the quality of the optically reconstructed patterns. Finally, the experimental results of the reconstructed patterns are in agreement with the theoretical results.

Fig. 10 Recorded patterns for 5G shape (a) and star shape (b) with GS algorithm

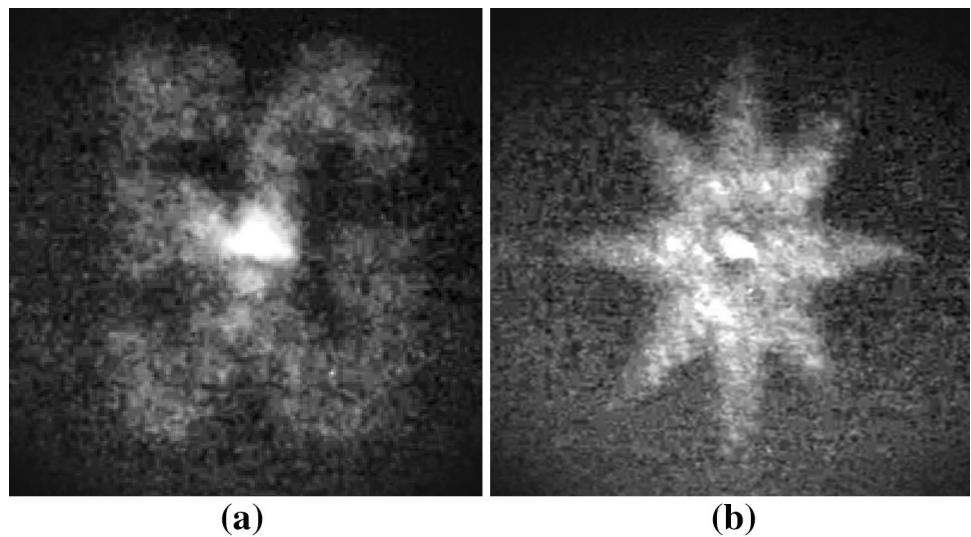


Fig. 11 Recorded patterns for 5G shape (a) and star shape (b) with MRAF algorithm

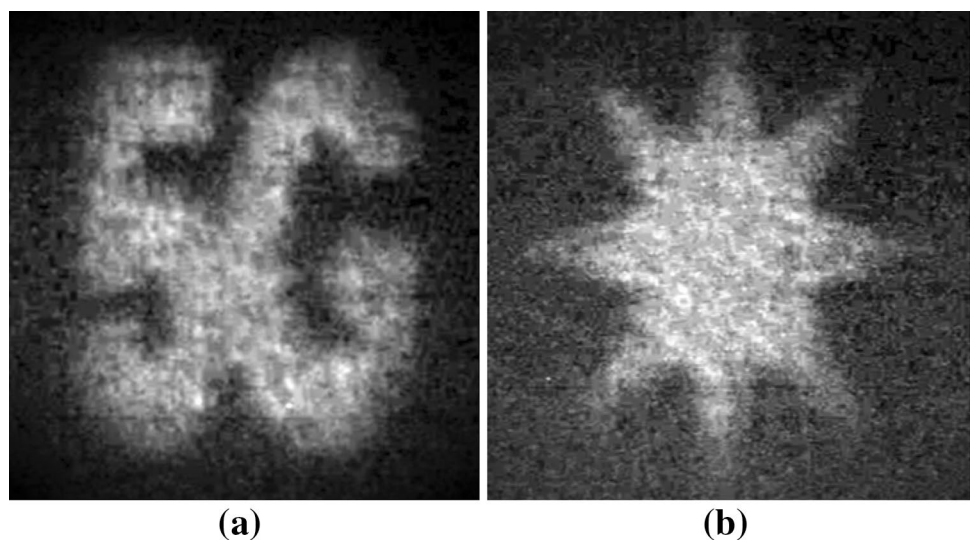


Table 2 Measured values of image quality of the experimental reconstructed patterns

Algorithm	Target patterns	MSE	PSNR	SC	NAE	NK	Cont	IDM
GS	5G shape	52.0857	30.9636	0.2297	0.6432	1.0018	0.2144	0.8939
	Star shape	68.2987	29.7867	0.3083	0.2607	1.0039	0.2954	0.8547
MRAF	5G shape	51.5745	31.0065	0.2334	0.4922	1.0006	0.1506	0.9247
	Star shape	36.6699	32.4877	0.3156	0.2062	1.0036	0.1766	0.9118

6.5 IQE of experimental results and discussion

Equations 15 up to 21 and the experimental results of recorded patterns are used to measure the image quality. IQE and identity of patterns are utilized to compare the experimental reconstructed patterns of the GS and MRAF algorithms. These comparisons are shown in Table 2. Table 2 is considered as the first comparison in the field of IQE utilizing these feedback algorithms.

From Table 2, *NK* indicates the proximity between the original patterns and the reconstructed patterns of these two test target patterns with the 5G shape using the MRAF algorithm. *NAE* values for the reconstructed patterns of the desired distribution obtained by the MRAF algorithm are smaller and more closely approach zero than their values obtained by the GS algorithm. The contrast values are low and *IDM* are high for the reconstructed patterns obtained by the MRAF algorithm, achieving their expected values, whereas for the GS algorithm, the contrast values are high and *IDM* are low when compared with the corresponding values of MRAF algorithm. The *SC* values of the MRAF algorithm are greater than those of the GS algorithm by a small amount. The *PSNR* values for the reconstructed patterns obtained by the MRAF algorithm are large compared with those obtained by the GS algorithm; a very clear star shape is observed for the MRAF algorithm case because of the low value of difference between the reconstructed patterns of star shape and the reference shape. Thus, we can conclude that the experimental reconstructed patterns obtained using the MRAF algorithm are better compared to those of the other algorithm. Thus, when comparing Tables 1 and 2, the theoretical and experimental results are consistent with each other, and image quality is improved when using the MRAF algorithm and a DOE, such as phase-only SLM.

SLM device. The MRAF algorithm was found to be better than the GS algorithm through the numerically and experimentally reconstructed optical patterns and also better than the method given by reference [30] in terms of reduced the unmodulated light beam (dc term) and the number of iterations. In experimental results, the optically reconstructed patterns by the MRAF algorithm also contained some distortion similar to the patterns of the GS algorithm, but with lower amplitude. The unmodulated light beam (dc term) was reduced in the patterns reconstructed by the MRAF algorithm without utilizing any suppressed method or by adding an additional tool in the optical setup. The reduction of the dc term from the patterns is based on the partitioning of the Fourier plane and the determination of the initial phase. Moreover, the GS and MRAF algorithms were compared in terms of the full reference objective quality and pixels homogeneity for the reconstructed patterns; the reconstructed patterns of the desired intensity distribution by MRAF algorithm were found to be enhanced and more bright. The theoretical and experimental results are consistent for both GS and MRAF algorithms. This comparison between GS and MRAF algorithms is important, because it can be adopted in many applications such as optical image processing, image encoding, optical communications, material processing, and in all fields which require smooth and homogenous intensity distribution in processing. Furthermore, from this comparison, it is concluded that the MRAF algorithm can be used to reduce the unmodulated light beam and achieve image quality enhancement with pixelated phase-only SLM.

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7 Conclusions

In this paper, two iterative phase-recapture algorithms based on the Fourier transform, namely, the Gerchberg–Saxton (GS) and mixed-region amplitude freedom (MRAF) algorithms, were studied. A detailed theoretical and experimental comparison was performed between these algorithms for implementing a digital hologram kinoform of two different desired intensity distributions based on the phase-only

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